De-Icing and Anti-Icing Simulation

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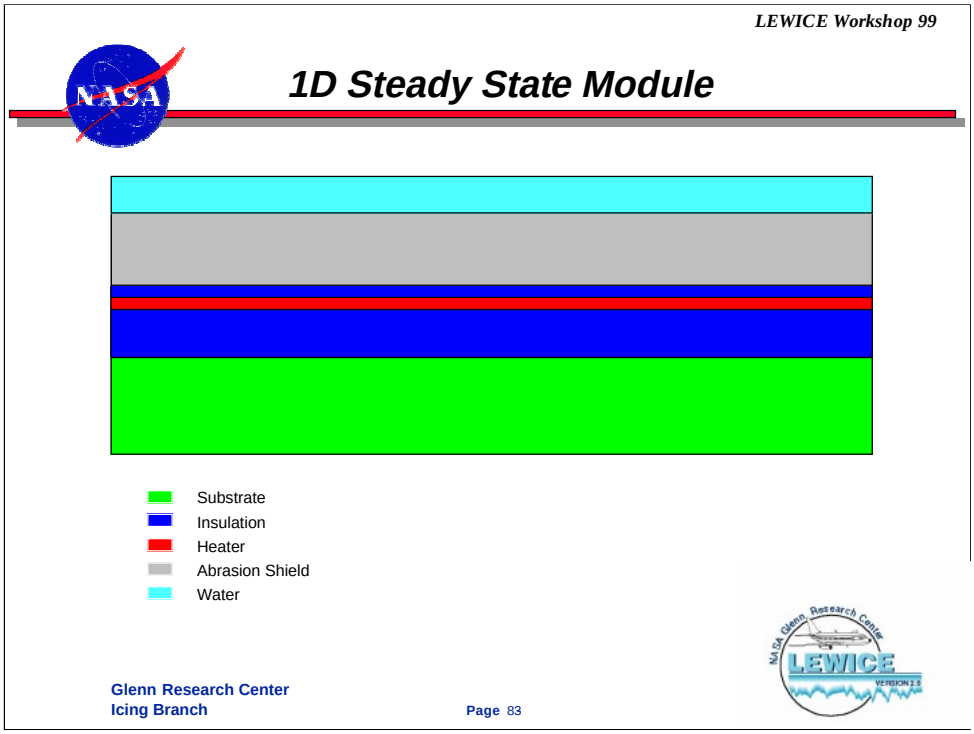
This presentation will discuss programs developed to simulate thermal deicing and anti-icing applications.

Codes Available

- **1D Steady state**
  - **included in LEWICE 2.0**
- **2D Anti-icing program (ANTICE)**
  - **Not interfaced with LEWICE**
    - » **In progress**
- **2D De-icing program (LEWICE/Thermal)**
  - **Interfaced with LEWICE 1.6**
  - **Interface with LEWICE 2.0 in progress**

Three codes are available for analyzing thermal systems. Each has their own region of specialization. The first module is built directly into LEWICE version 2.0. It is very useful for preliminary design studies but does not provide detailed analysis. The second code, ANTICE, is used to more thoroughly analyze anti-icing systems, both hot air and electrothermal. The third option, LEWICE/Thermal, is used to simulate electrothermal deicers.



This diagram shows the anti-icing system modeled in LEWICE. It consists of a 1D heater at each chordwise location. The model shown is an electrothermal system, but hot air systems can also be analyzed.

1D Steady State Module

- Anti-icing only
  - Running wet or evaporative
  - Hot air or electrothermal
- No axial conduction
- No anisotropic materials
  - Composites
- Simple, fast
- Input desired surface temperature
- Outputs heat requirements
- Heater temperature prediction unrealistic

The 1D anti-icing model in LEWICE can simulate running wet or evaporative modes and hot air or electrothermal systems for a total of 4 possible options. To obtain a fast solution, it neglects chordwise and spanwise conduction and cannot model anisotropic materials. The user inputs a desired surface temperature and the module outputs the heat required to maintain that surface temperature. Other inputs are the mass flow rate (for hot air systems) and the thermal properties and thicknesses of the materials.

This module was designed to answer preliminary design questions such as:

How much heat do I need to maintain a surface temperature of ___?

How much heat do I need to completely evaporate the water?

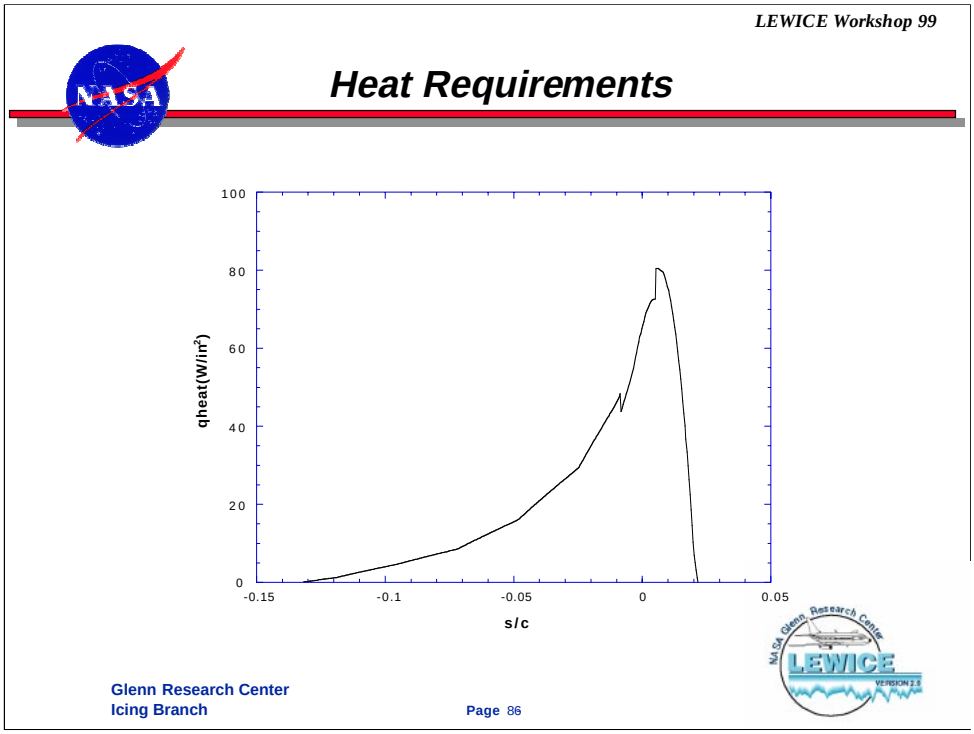
In an evaporative mode, the surface temperature is also calculated from the energy balance.

The useful outputs from this module are the surface temperature and heat requirements. Do to the simplistic model, heater temperatures or hot air temperatures are not predicted well.

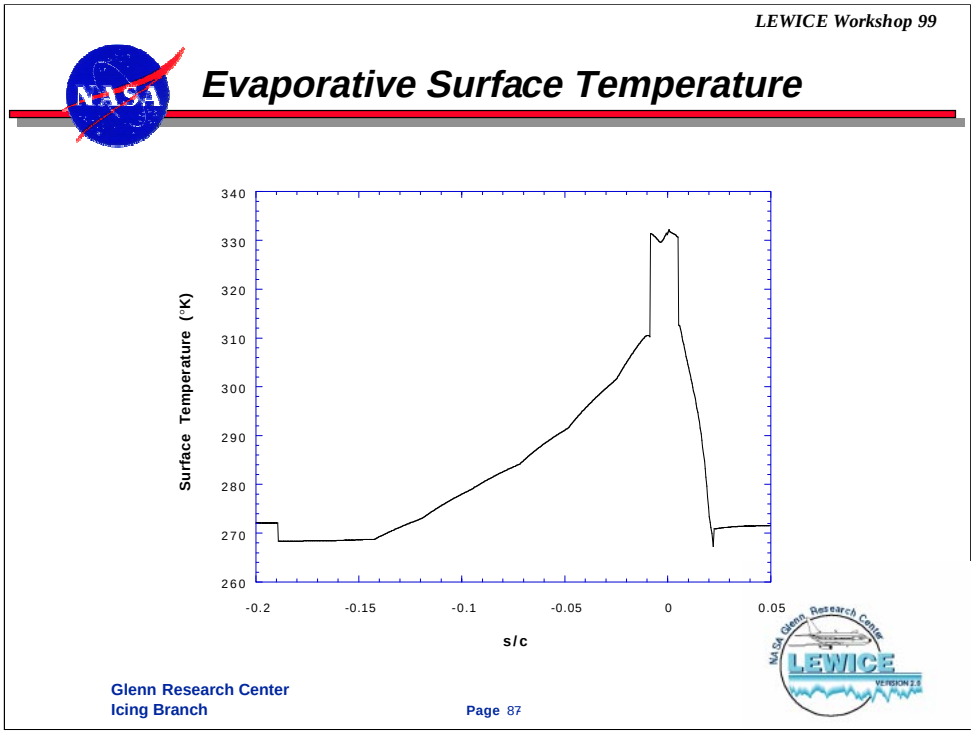
Sample Case

- **Example 9 from user manual**
- **MVD = 20 μm**
- **LWC = 0.54 g/m^3**
- **T (static) = -5 ˚C**
- **A = 4.5˚**
- **Chord = 36 inches**
- **V = 90 m/s = 175 kts**

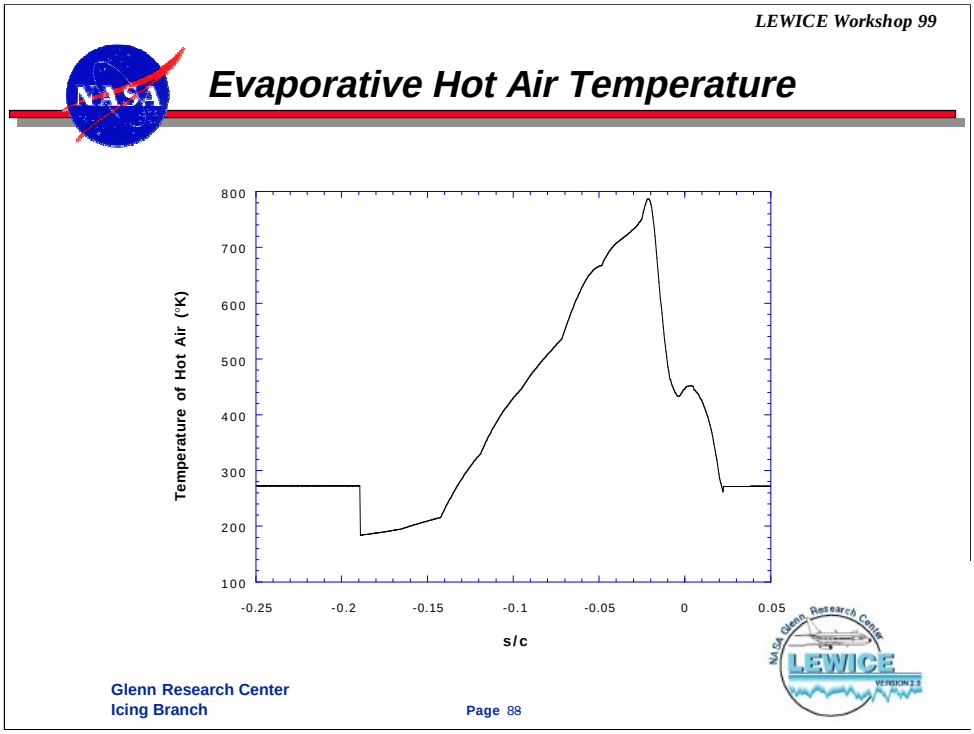
This is an example case used to illustrate the output from an evaporative hot air anti-icing system. The case shown is Example Case 9 in the User Manual.



This plot shows the amount of heat needed to completely evaporate the water for the conditions given. Because of the high heat requirements, many systems are designed as running wet instead of evaporative. Additionally, since there will be conduction chordwise in this system, many users will average the heat requirements given above over the impingement region.



This plot shows the predicted surface temperature needed to evaporate the incoming model. This temperature is dependant only on the outer airfoil geometry and conditions. It does not depend upon the material properties or thicknesses.



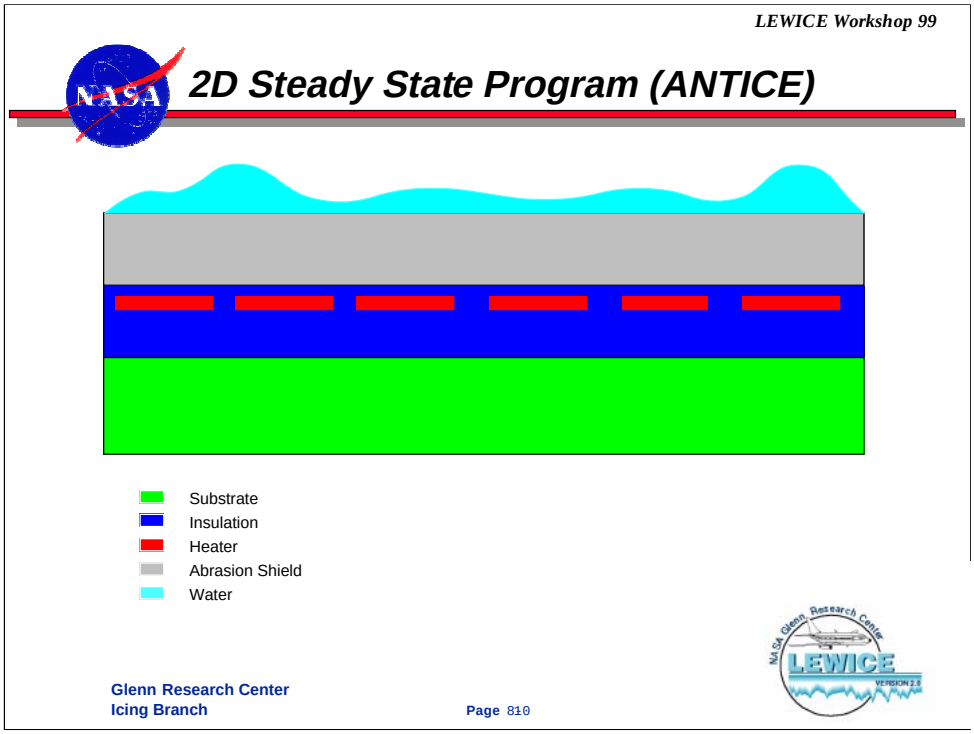
This plot shows the predicted temperature of the hot air stream which is needed to maintain evaporative conditions. This temperature can be more accurately predicted with the ANTICE code.

ANTICE

- **Flow solution input**
- **Collection efficiency input**
- **File format compatible with LEWICE**
- **Anti-icing only**
- **Future integration with LEWICE 2.0**

ANTICE is a stand-alone code (not tied into LEWICE) which analyzes hot air or electrothermal anti-icing systems. The flow solution, collection efficiencies, and heat transfer coefficients are read in as inputs. The file format is compatible with the output files written from LEWICE. The system takes a given heat input (which can be estimated using the previous module) and outputs the temperature distribution within the anti-icer and the water.

This code will be incorporated into LEWICE in future versions.



This diagram shows a representation of the anti-icing simulation performed by ANTICE. The system shown is an electrothermal model, but hot air systems can also be modeled

ANTICE Capabilities

- **Multi-layer structure**
- **Anisotropic materials**
- **Hot air or electrothermal**
- **Multi-zone, variable wattage thermal heaters**
- **Determines if and where film breaks into rivulets**
- **Calculates runback, evaporation and freezing rate**
- **Calculates hot air temperature**

ANTICE can model a multi-layered composite structure with anisotropic materials. The heat can be supplied by electrothermal or hot air devices. Each heater is individually controlled and the hot air is supplied via input of the inside surface heat transfer coefficients. It models the surface water runback and determines where and if rivulets form. The hot air temperature as well as other temperatures are predicted by the model.

ANTICE Inputs and Outputs

- **Main input file**
  - **Included thicknesses and thermal properties of materials**
- **FLOW.DAT, BETA.DAT, HTC.DAT from LEWICE**
- **HI.DAT**
  - **Internal heat transfer coefficients**
- **STREAM.DAT**
  - **Streamline information for pseudo-3D simulation**
- **Outputs temperature distribution**

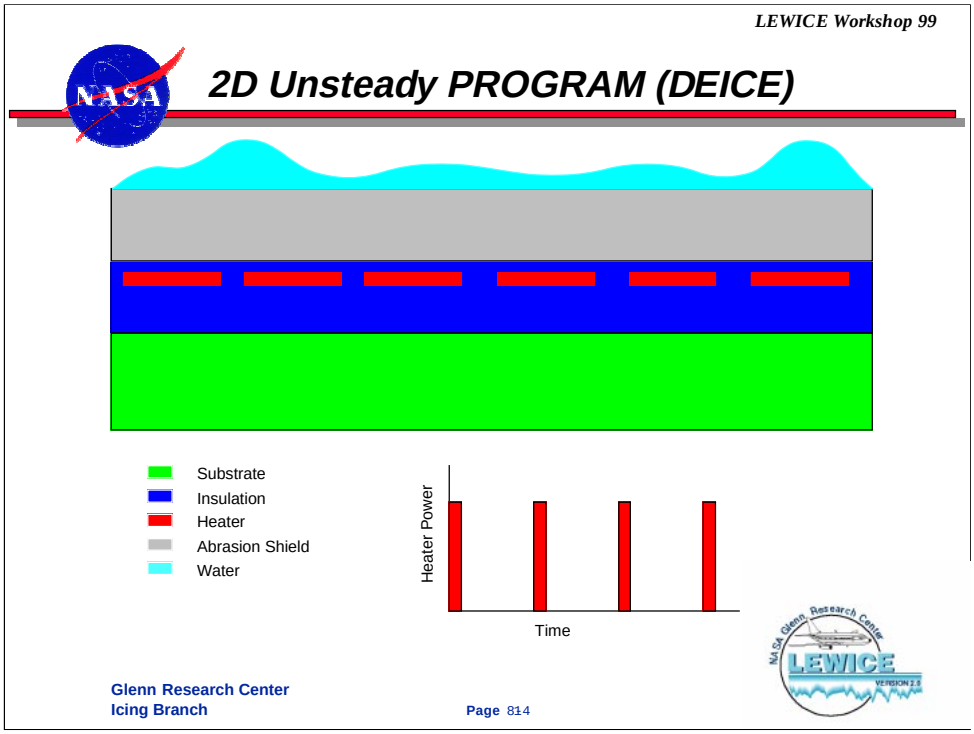
The input files for ANTICE are the output files from LEWICE, a main input file, and input files for the hot air heat transfer coefficients and streamline data. The main input file includes thermal properties, thicknesses and widths of the different materials. It also includes the hot air mass flow rate and location of the air supply. The streamline input file can be used to simulate a pseudo-3D heat transfer by supplying the distances between two span wise streamlines. The outside heat transfer coefficients can be calculated by ANTICE or by supplied from LEWICE.

ANTICE outputs the temperature at each location in the hot air stream, the airfoil materials and the water. It can predict an ice thickness but does not modify the surface geometry as does LEWICE.

LEWICE/Thermal

- **Rectangular deicer model**
  - **Curvature effects embedded in flow solution and collection efficiencies**
- **Transient, 2D Model**
  - **Accretion**
  - **Melting**
  - **Runback**
  - **Ice Shedding**
  - **Refreezing**

The third thermal option is LEWICE/Thermal, which models an electrothermal deicer system. Since the heaters can turn on and off, there can be ice buildup, melting and shedding of ice, surface water runback and refreezing of surface water. The deicer model itself is rectangular, as the curvature effects are embedded in the flow solution and the collection efficiency.



This diagram illustrates an example of an electrothermal deicer system modeled by LEWICE/Thermal. The plot in the corner shows a representative heater cycle time.

LEWICE/Thermal Features

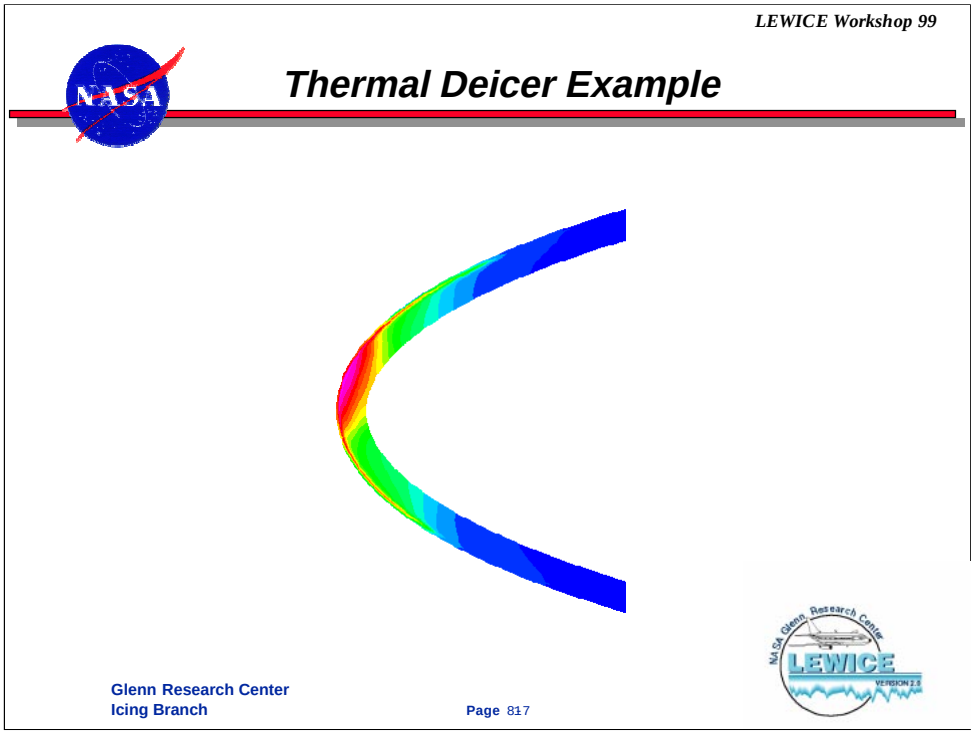
- **Multi-layered composite bodies**
- **Anisotropic materials**
- **Independent heater power and cycle times**
- **Can model gaps in heaters**
- **Ice shedding prediction**
- **Melting and refreezing**
- **Runback modeling**

This program can model a multi-layered composite deicer with anisotropic materials. Each heater is independently controlled with its own power and cycle time. As stated earlier, it can model melting and shedding of ice, water runback and refreezing of runback water including rivulet formation.

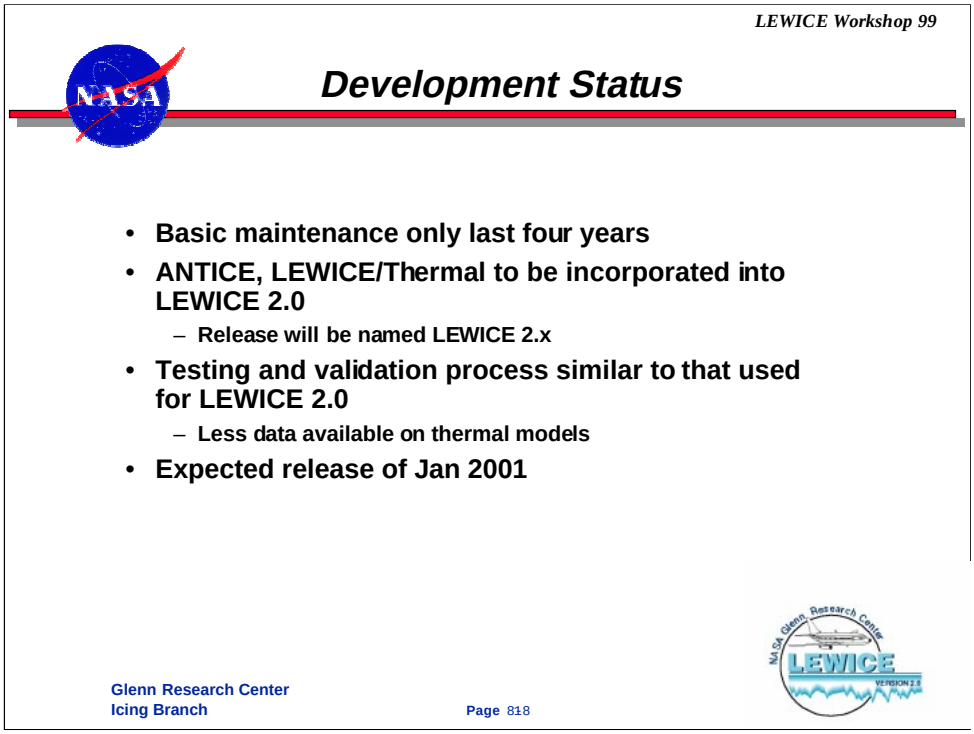
LEWICE/Thermal Input and Output

- **Input material thickness and properties**
- **Input heater power density and cycle times**
- **Outputs columns of temperatures at user-selected locations**
- **Outputs PLOT3D format file for contour plots**

Inputs for LEWICE/Thermal include material thicknesses and thermal properties, heater wattages and cycle times as well as the standard LEWICE input file. It can output temperatures at selected locations, representative of output from a thermocouple and/or it can output the complete temperature distribution at user-selected times in a PLOT3D compatible format. This is useful for creating thermal contour plots. Since PLOT3D is a flow visualization package, it outputs temperature in place of air density as the scalar variable. Other data needed for the PLOT3D format contains dummy information.



This plot shows an example thermal contour for a deicer system. The plot shows a close up of the leading edge of an airfoil with an electrothermal system. In the plot, the parting strip heater at the leading edge is on, showing the thermal contours within the airfoil. No ice is shown in the figure, although that can be represented as well.



Both ANTICE and LEWICE/Thermal have been supported with basic maintenance the last four years. The LEWICE/Thermal model was added to LEWICE version 1.6 but was otherwise unmodified.

Work is underway to incorporate both ANTICE and LEWICE/Thermal into LEWICE, with the resulting code released as LEWICE 2.x (probably named version 2.1 although that is undecided).

There will be a thorough testing and validation process performed next year for this code, similar to the validation effort undertaken with LEWICE 2.0. However, there is less data publicly available for validation of these models.

The expected release date for this version will be January 2001 with a presentation at that year's AIAA conference in Reno, NV.

Discussion

- **Desired features**
- **Input/Output formats**
- **Links to commercial codes**
- **Development priority relative to LEWICE and other codes**

Since development is underway on these models, this is an excellent opportunity to provide feedback on these models.

What features are desired?

Would you like a more user-friendly way to input deicer geometries?

(Be specific)

Would it be more advantageous to interface these models to commercial heat transfer codes? Which ones?

Due to limited manpower, what is the priority of these codes compared to LEWICE or other codes?